

Computer Networks Spring 2026

Assignment#4 (6A & 6B)

Due Date: Thursday, April 2nd, 2026

Submission Mode & Time: Submit a PDF of your scanned handwritten solution online (on Google Classroom) if classes remain online until the due date. If classes are held in person, submit handwritten solutions during the lecture.

Please note the following:

1. If classes are online on the due date (2nd April 2026), submit a PDF of your scanned handwritten solutions through the Classwork Section on Google Classroom.
2. If classes are in-person (physical) on the due date (2nd April 2026), bring and submit your handwritten solutions physically during the lecture.
3. Assignment document emailed to the Course Instructor or the TA or assignments submitted after the due time will not be accepted.
4. No exceptions to the above date and time will be allowed. Inability to submit the assignment by the required time will result in zero marks.
5. To ensure self-completion of assignments and discourage plagiarism, the instructor or the relevant TA may randomly contact you and ask for an explanation of your answers. Where plagiarism and/or cheating is evident, you will be referred to the departmental disciplinary committee. In extreme cases of plagiarism an F may be awarded immediately with further referral to the university disciplinary committee.
6. All solutions must be **hand-written**.
7. While collaboration in this course is highly encouraged, you must ensure that you do not claim other people's work/ ideas as your own.

Question: 01 [5 Marks] [CLO 1]

Consider two IPv6 nodes (A and B) communicating over an IPv4-only core network using an IPv6-over-IPv4 tunnel.

If Node A sends a standard IPv6 datagram with a payload of 1000 bytes (assuming a standard 40-byte IPv6 header and no extension headers), what is the total size of the IP packet (in bytes) that physically traverses the IPv4 wire? Explain the calculation.

Question: 02 [4 Marks] [CLO 1]

Why is Network Address Translation (NAT) often described as violating the "end-to-end" principle of the Internet architecture?

Question: 03 [4 Marks] [CLO 1]

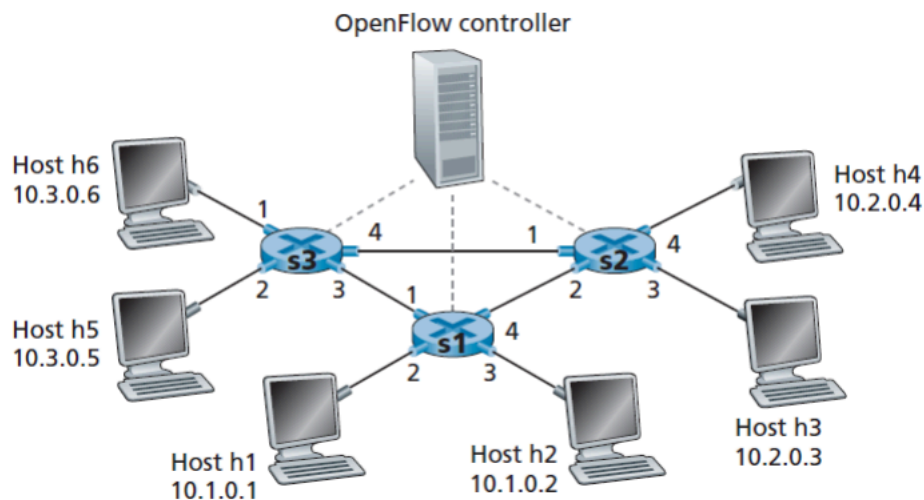
If a user behind a NAT router runs a P2P application (like BitTorrent) and wants to accept incoming connections from other peers, why does this fail by default? What specific manual configuration is usually required on the router to fix this?

Question: 04 [5 Marks] [CLO 1]

IPv6 has a much larger address space (128-bit) than IPv4 (32-bit), which naturally increases the header size. Name two fields found in the IPv4 header that were *removed* or *moved to extension headers* in IPv6 to keep processing efficient.

Question: 05 [2+2+2+3= 9 Marks] [CLO 3]

Given a network with 6 hosts (h1 to h6) and three packet switches (s1, s2, and s3), each with four interfaces, and each with four local interfaces (numbered 1 through 4). We'll consider a number of network-wide behaviors that we would like to implement, and the flow table entries in s1, s2 and s3 are needed to implement this behavior.



- What specific flow table entry should be added to switch S1 to block all traffic from Host h1 to Host h4? State (provide) the exact match conditions and action.
- State (provide) the exact match condition and action in switch S1 which is required to allow only HTTP (port 80) and SMTP (port 25) traffic from (Host h1 or Host h2) to Host h5 via interface 1.
- State (provide) the exact match condition and action required to drop all other traffic (except http) from h2/h1 to h5 considering the previous scenario given in part B.
- Specify the flow table entry in switch S3, S2 and S1 to forward all traffic from Host h5 to Host h1 towards S2 and then S2 forwards these packets to S1. Suppose S3->S2->S1 is the only possible path from h5 to h1. State (provide) the exact match conditions and action.

Question: 06 [14 Marks] [CLO 2]

Host A sends a single IPv4 datagram to Host B over the Internet. There is a single network path from A to B with two intermediate routers (R1 and R2). Original datagram exiting Host A has following values in its IPv4 header:

Total Length (Bytes)	IP Header Length (Bytes)	Identification (ID) (Hexadecimal)	DF Flag	MF Flag	Fragment Offset
1600	20	0xCAFE	0	0	0

Note that while a real IPv4 header field of *IP Header Length* counts words of 4 Bytes, we have used the units of Bytes for simplicity. Additionally note that no IP options are used. Recall that *fragment offset* field is 13 bits wide in IPv4 header. Moreover, DF stands for Don't fragment while MF stands for More Fragments.

Maximum Transmission Unit (MTU) on the link connecting Host A to Router R1 is 9000 Bytes.

MTU on the link connecting router R1 to router R2 is 1500 Bytes.

MTU on the link connecting router R2 to the destination Host B = 512 Bytes.

- Draw a 6-column table (like the example above) and fill in the details for every fragment created by router R1.
- Draw the same 6-column table. Use it to list all sub-fragments (if any) created by router R2 as it processes the packet(s) from R1.

Question: 07 [4*5 + 3 = 23 Marks] [CLO 3]

You are given the network 192.168.48.0/22 for an organization with the following subnet requirements:

- CS Department requires 100 hosts.
- IT Department requires 62 hosts.
- Finance Department requires 50 hosts.
- Admin office requires 20 hosts.
- Director office requires 10 hosts.

Your task is to allocate addresses optimally (as per given requirements) performing VLSM as per the given requirements, determine the following for each subnet and complete the table given below (**please provide your final answer in dotted decimal notation only**):

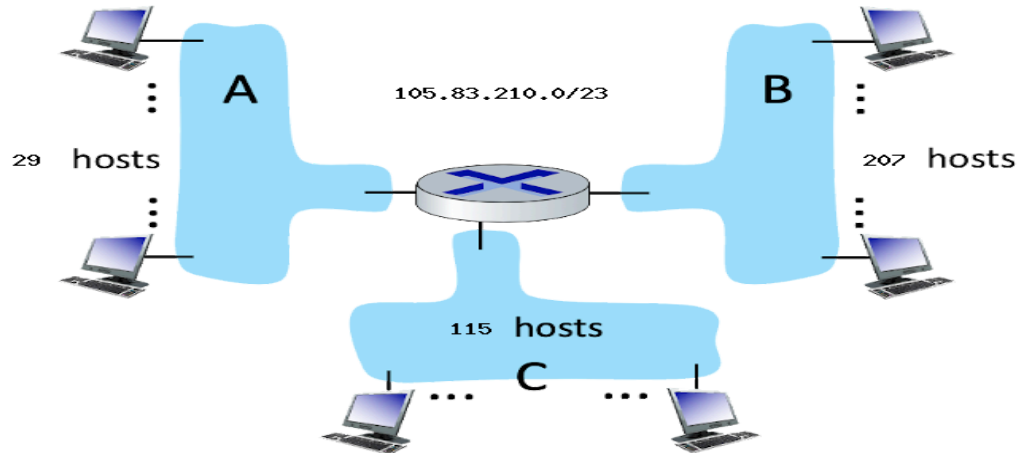
- Prefix / Subnet Mask
- Network Address
- First Usable Host Address
- Broadcast Address

Moreover, find the leftover / free addresses inside /22 after fulfilling the above requirement optimally.

Department /Office	Subnet Mask	Network Address	First Usable Host Address	Broadcast Address	No. of free addresses
CS					
IT					
Finance					
Admin					
Director					
No. of Free addresses					

Question: 08 [1*10 = 10 Marks] [CLO 2]

Refer to the figure below, three subnets need to be assigned to subnets A, B & C from the available parent IP address block 105.83.210.0/23 in a manner so that assigned IP addresses are minimal, leaving the largest amount of contiguous space available for future subnets. Please answer the following questions: (.Please provide all final answers in dotted decimal format CIDR notation.



- i. Maximum number of hosts in the parent IP address block:
- ii. Subnet Address of Subnet A (in CIDR notation):
- iii. First assignable IP in Subnet A (in CIDR notation):
- iv. Broadcast address of Subnet A (in CIDR notation):
- v. Subnet Address of Subnet B (in CIDR notation):
- vi. First assignable IP in Subnet B (in CIDR notation):
- vii. Last assignable address of Subnet B (in CIDR notation):
- viii. Subnet Address of Subnet C (in CIDR notation):
- ix. Last assignable IP in Subnet C (in CIDR notation):
- x. Broadcast address of Subnet C (in CIDR notation):

Question: 09 [2+4.5+4.5 = 11 Marks] [CLO 3]

An ISP is granted a block of addresses starting with 170.25.96.0/20. The ISP needs to distribute these addresses to three groups of customers as follows:

- The first group has 8 customers: each customer needs 256 addresses (including network and broadcast).
- The second group has 16 customers: each customer needs 64 addresses (including network and broadcast).
- The third group has 64 customers: each customer needs 16 addresses (including network and broadcast).

The IT department of ISP designs sub-block (subnets) for each group of customers strictly according to addresses required by customers in each group.

You are required to answer the following questions with respect to this scenario. Make sure that IP addresses are necessarily written in dotted decimal notation.

1. Write the subnet mask for each group of customers.
2. Write the network address for 1st customer of each group:
3. Write the broadcast address of last customer of each group: